Georgios Liargkovas

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Education

Columbia University, Department of Computer Science — *New York, NY, USA*
PhD in Computer Science — *Sep 2024 - Present*
- **Advisor:** Prof. Kostis Kaffes
- **Anticipated graduation:** May 2029

Columbia University, Department of Computer Science — *New York, NY, USA*
M.S. in Computer Science — *Sep 2024 - Dec 2025*
- **GPA:** 3.95/4

Athens University of Economics and Business, Department of Management Science & Technology — *Athens, Greece*
BSc in Management Science & Technology — *Sep 2018 - Mar 2023*
- **GPA:** 9.37/10 (2nd among 120 students – top 1.6%)
- **Major:** Software Engineering and Data Analysis

Research and Work Experience

Google AI2, Prodkernel, Memory Management Team — *Sunnyvale, CA, USA*
Student Researcher — *May 2026 - Aug 2026*
- Agentic development workflows for Linux kernel tuning and optimization using large language models.

Brown University — *Providence, RI, United States*
Visiting Research Fellow — *Oct 2022 - May 2024*
- **Advisor:** Prof. Nikos Vasilakis
- Created `hs`, a speculative, out-of-order execution engine for POSIX shell scripts.
- Co-developed `try`, a lightweight tool that lets users run a command and inspect its effects before changing their live system, using Linux namespaces and the OverlayFS union filesystem (5.5k+ GitHub stars).
- **Technologies used:** Python, Shell scripting (bash/zsh), Docker, System call tracing & isolation, Linux namespaces, OverlayFS

Athens University of Economics and Business, BALab, AUEB — *Athens, Greece*
Junior Researcher — *Mar 2020 - Aug 2024*
- **Advisor:** Prof. Diomidis Spinellis
- Carried out research about the gap between software practitioners' education and industry demands.
- Examined the use of static analysis warning suppression to detect motives behind its use and impact on software.

Secure Open Source Software and Hardware Adaptable Framework (SecOPERA) — *Athens, Greece*
Research Software Engineer — *May 2023 - May 2024*
- **Advisor:** Prof. Panos Louridas
- Produced requirement specifications and contributed to the architecture of SecOPERA, an EU Horizon Europe project for secure open-source software and hardware solutions (URL).

Publications

Google Scholar: 72 citations, h-index 4 (July 14, 2026)
** denotes equal contribution*

[SOSP '26] **Georgios Liargkovas**, Mihir Nitin Joshi, Hubertus Franke, and Kostis Kaffes. "TuxBot: Semantic-Aware Online OS Tuning with Large Language Models." In *ACM Symposium on Operating Systems Principles (SOSP '26)*. To appear, 2026.

[OSDI '26] **Georgios Liargkovas***, Di Jin*, Tianyu (Ezri) Zhu, Dan Liu, A. Bolun Thompson, Anirudh Narsipur, Seong-Heon Jung, Siddhartha Prasad, Diomidis Spinellis, Michael Greenberg, Konstantinos Kallas, and Nikos Vasilakis. "hS: Speculative Script Reordering at Subprocess Granularity." In *USENIX Symposium on Operating Systems Design and Implementation (OSDI '26)*, July 2026.

[OSDI '26] Evangelos Lamprou*, Tianyu (Ezri) Zhu*, Di Jin, Grigoris Ntousakis, **Georgios Liargkovas**, Calvin Eng, Konstantinos Kallas, Michael Greenberg, and Nikos Vasilakis. "Controlling Opaque-Component Effects with Semisolates and Try." In *USENIX Symposium on Operating Systems Design and Implementation (OSDI '26)*, July 2026.
🏆 Best Paper, Distinguished Artifact.

[ICLR '26] Naimeng Ye*, Arnav Ahuja*, **Georgios Liargkovas***, Yunan Lu*, Kostis Kaffes, and Tianyi Peng. "Speculative Actions: A Lossless Framework for Faster AI Agents." In *International Conference on Learning Representations (ICLR '26)*, 2026.
🏆 Oral.

[IEEE DEB '26] Elaine Ang, Chenxi Huang, **Georgios Liargkovas**, Jerry Liu, Jinhui Liu, Nikos Pagonas, Charlie Summers, Haonan Wang, Jiakai Xu, Tianle Zhou, Yusen Zhang, Zhou Yu, Zhuo Zhang, Tianyi Peng, Kostis Kaffes, and Eugene Wu. "Agentic Data Environments." *IEEE Data Engineering Bulletin*, March 2026.

[ML4Sys '25] **Georgios Liargkovas**, Vahab Jabrayilov, Hubertus Franke, and Kostis Kaffes. "An Expert in Residence: LLM Agents for Always-On Operating System Tuning." In *ML for Systems Workshop at NeurIPS*, 2025.

[PACMI '25] **Georgios Liargkovas**, Prabhpreet Singh Sodhi, and Kostis Kaffes. "Set It and Forget It: Zero-Mod ML Magic for Linux Tuning." In *Proceedings of the 4th Workshop on Practical Adoption Challenges of Machine Learning for Systems (PACMI '25)*, 2025.

[eBPF '25] Prabhpreet Singh Sodhi, **Georgios Liargkovas**, and Kostis Kaffes. "Empowering Machine-Learning Assisted Kernel Decisions with eBPFML." In *Proceedings of the 3rd Workshop on eBPF and Kernel Extensions (eBPF '25)*, 2025.

[HotOS '23] **Georgios Liargkovas**, Konstantinos Kallas, Michael Greenberg, and Nikos Vasilakis. "Executing Shell Scripts in the Wrong Order, Correctly." In *Proceedings of the 19th Workshop on Hot Topics in Operating Systems (HotOS '23)*, 2023.

[IEEE TE '21] **Georgios Liargkovas**, Angeliki Papadopoulou, Zoe Kotti, and Diomidis Spinellis. "Software Engineering Education: Knowledge Versus Industrial Needs." *IEEE Transactions on Education*, 2021.

Teaching Experience

Columbia University, Department of Computer Science — *New York, NY, USA*
COMS 9902: Frontiers in CS Research, **Teaching Assistant (TA)** — *Spring, 2026*
- PhD-level course with 30 students.

Athens University of Economics and Business — *Athens, Greece*
Programming II, **Teaching Assistant (TA)** — *Fall 2023*
- Undergraduate course with 180 students; delivered bi-weekly lectures on Compilers, Data Structures, Functional and Generic Programming in Java.
- Aided students with course assignments during weekly lab support sessions.

Athens University of Economics and Business — *Athens, Greece*
Applied Software Engineering, **Teaching Assistant (TA)** — *Spring 2022, 2023*
- Undergraduate course with 50 students; evaluated assignments and advised students on coursework.

Athens University of Economics and Business — *Athens, Greece*
Introduction to Computer Science, **Head Teaching Assistant (Head TA)** — *Fall 2022*
- Head TA for an undergraduate course with 180 students; coordinated course organization, lectures, labs, assignments, and final assessment.
- Lectured weekly on Computer Architecture, Networks, Algorithms, Data Structures, and Databases.
- Fully automated the final examination by creating a customizable multiple-choice exam generator.

Service

Reviewing & PC service: AdaptFM 2026 (*PC member*); SOSP 2026 (*Artifact Evaluation Committee*); EuroSys 2026 (*Artifact Evaluation Committee*); USENIX ATC 2025 (*Artifact Evaluation Committee*); EuroSys 2024 (*Shadow PC*).
Conference service: MSR 2023 (*Web Chair*); ICSE 2022 (*Student Volunteer*); ESEC/FSE 2022 (*Student Volunteer*).
Open source: JabRef (*Contributor*, PR #7846).

Honors and Awards

2025	**Dreamsports AI Grant,** Dreamsports AI Innovation Center, Columbia University	*New York, NY, USA*
2024	**SEAS Doctoral Fellowship,** Columbia University	*New York, NY, USA*
2023	**HotOS Student Grant,** ACM SIGOPS	*Providence, RI, USA*
2022	**Research Scholarship Award,** Athens University of Economics and Business	*Athens, Greece*
2022	**ACM SIGSOFT CAPS Grant,** ICSE 2022	*Pittsburgh, PA, USA*
2022	**ACM SIGSOFT Support Grant,** Student Mentoring Workshop, ICSE 2022	*Pittsburgh, PA, USA*
2022	**Honorary Award,** Athens University of Economics and Business	*Athens, Greece*